Testing of cement and types of cement

Notes from the lecture slides "Module 1c – Testing and Types of cement"

In this module

1. Physical tests on cement: overview
2. Fineness of cement
3. Standard consistency
4. Setting time
5. Soundness
6. Strength of cement
7. Types of cement

1 Physical tests on cement: overview

Property	Why it matters	Test (Indian Standard)
Fineness	Controls the rate of hydration and strength gain	Blaine air permeability, IS 4031 (Part 2)
Standard consistency	Fixes the water content used for the other paste tests	Vicat apparatus with 10 mm plunger, IS 4031 (Part 4)
Setting time (initial and final)	Time available for mixing, transporting and placing; time before the concrete can carry load	Vicat needles, IS 4031 (Part 5)
Soundness	Cement must not expand after hardening	Le Chatelier and autoclave tests, IS 4031 (Part 3)
Compressive strength	Grade of the cement	70.6 mm mortar cubes, IS 4031 (Part 6)

2 Fineness of cement

2.1 Why fineness matters

- Hydration starts at the **surface** of the cement grains, so the **surface area decides how much material is available to react**.
- **Finer cement → faster hydration → faster strength development** (and more early heat).
- But grinding finer costs more energy and money, and very fine cement shrinks more and needs more water, so **cost-effectiveness** must be considered.
- Fineness is expressed as **specific surface in m^2/kg** (surface area of 1 kg of cement).

Cement (IS 269:2015)	Minimum specific surface
OPC 33, 43 and 53	**225 m^2/kg**
OPC 43S and 53S (special grades for railway sleepers)	**370 m^2/kg**

2.2 Particle size distribution methods

A direct way to find specific surface is to measure the **particle size distribution** by **sedimentation or elutriation**.

- These rely on **Stokes' law**: a particle falling through a fluid reaches a **terminal velocity** that depends on its size, so particles of different sizes settle at different rates.
- **Wagner turbidimeter** (ASTM C115): cement is suspended in **kerosene**. A **beam of light** passes through the suspension at a known level, and a **photocell** measures how much light gets through. Less light means more particle area in the beam, which gives the concentration of particles of each size and hence the surface area.

2.3 Air permeability methods (used today)

Lea and Nurse method (BS 4550 Part 3)

- Dry air flows at a **constant velocity** through a bed of cement of **known porosity and thickness**.

- The **pressure drop** across the bed is measured. Finer cement has more surface area, so it resists the air flow more.
- The specific surface is calculated from the **permeability of the bed**.

Blaine air permeability method (ASTM C204; IS 4031 Part 2)

- Here the air does **not** flow at a constant rate. A **known volume of air** is drawn through a compacted bed of cement at a prescribed average pressure, and its flow rate falls steadily.
- The **time taken** for this volume of air to pass is measured. A longer time means a finer cement.
- For a given apparatus and a standard porosity, the specific surface is calculated:

$$S = \frac{K}{\rho} \times \frac{\sqrt{(e^3)}}{1 - e} \times \frac{\sqrt{t}}{\sqrt{(0.1\,\eta)}} \quad (cm^2/g)$$

where K = apparatus constant (found by calibrating with a standard cement), ρ = density of cement, e = porosity of the bed, t = time of flow (s), η = viscosity of air.

> **Typical short answer**
>
> *Why is the Blaine method preferred to sieving?* Sieving only tells you the fraction of large particles; it says nothing about the very fine particles that control early hydration. The Blaine test measures the **surface area**, which is what actually governs the rate of reaction.

3 Standard consistency

The **standard (normal) consistency** is the **water content that gives a paste of a standard stiffness**. It is found first, because the setting time and soundness tests must be done on paste of standard consistency, so that results from different cements can be compared.

Test (IS 4031 Part 4) using the Vicat apparatus:

1. Mix cement with a trial percentage of water and fill the Vicat mould.

2. Lower the **10 mm diameter, 50 mm long plunger** gently onto the paste and release it to sink **under its own weight**.
3. Repeat with different water contents until the plunger **stops 5 to 7 mm from the bottom of the mould**.
4. That water content, **P**, as a percentage of the dry cement mass, is the standard consistency.

- Typical values: **26–33%** of the mass of dry cement.

Vicat attachment	Size	Used for
Plunger	10 mm dia., 50 mm long	Standard consistency
Initial set needle	1 mm square, 50 mm long	Initial setting time
Final set needle	1 mm square needle with a 5 mm dia. annular (ring) attachment; the needle projects 0.5 mm beyond the ring	Final setting time

The mould is a split ring (about 80 mm across and 40 mm deep) on a non-porous plate. The moving rod carries an indicator over a scale.

4 Setting time

4.1 What setting is

Setting is the **stiffening** of the paste, from a fluid to a rigid state. It is different from *hardening*, which is the gain of strength after setting.

- Setting is caused mainly by the **selective hydration of C_3A and C_3S**, and it is accompanied by a **temperature rise** in the paste.
- **Initial set** corresponds to the **rapid rise** in temperature.
- **Final set** corresponds to the **peak temperature**.

4.2 False set and flash set

	False set	Flash set
When	**Within a few minutes** of adding water	Immediately on mixing
Heat	Little heat	**Large heat** release
Cause	Gypsum dehydrated to hemihydrate during hot grinding; it rehydrates and stiffens the paste	**No (or too little) gypsum**: C_3A reacts instantly
Remedy	**Remix** the paste without adding water; it regains plasticity and sets normally	Cannot be remixed; strength is affected

4.3 Initial setting time (IS 4031 Part 5)

1. Make a paste with **0.85 P** water (P = standard consistency).
2. Lower the **1 mm square needle** onto the paste and release it at intervals.
3. **Initial setting time** = time from **adding the water** until the needle **fails to penetrate beyond 5 ± 0.5 mm from the bottom** of the mould.

Minimum initial setting time (IS 269:2015):

- OPC 33, 43, 53: **30 minutes**
- OPC 43S and 53S: **60 minutes**

4.4 Final setting time

- Use the **needle with the hollowed-out metal attachment** (annular collar).
- **Final setting time** = time from adding water until the needle **makes an impression on the paste surface but the cutting edge of the attachment fails to do so**.
- **Maximum 10 hours (600 min)** for Portland cements.
- An approximate relation between the two: **final setting time (min) ≈ 90 + 1.2 × initial setting time (min)**.

> **Remember**
>
> Initial set must be **long enough** to mix, transport, place and compact the concrete. Final set must be **short enough** that formwork can be removed and work can continue. That is why the code gives a **minimum** for initial set and a **maximum** for final set.

5 Soundness

5.1 What unsoundness is

A **sound** cement does not undergo a large change in volume after it has hardened. An **unsound** cement **expands** after setting, and under restraint this expansion can **crack and disrupt** the hardened paste.

Unsoundness is caused by **delayed hydration** of three things:

1. **Free lime (CaO)**: lime that did not combine in the kiln, intercrystallized with other compounds. It **hydrates slowly** after the paste has hardened, and $Ca(OH)_2$ occupies a **larger volume** than the CaO it came from. (A chemical analysis cannot tell unreacted CaO from $Ca(OH)_2$, so a physical test is needed.)
2. **Magnesia (MgO)**: reacts with water like CaO, but much more slowly; its crystalline form (periclase) is **deleteriously reactive**.
3. **Calcium sulfate (excess gypsum)**: gypsum not used up by C_3A during setting reacts later to form **expansive ettringite** (calcium sulphoaluminate).

5.2 Le Chatelier test (accelerated test)

Detects unsoundness **due to free lime only**.

1. Fill the **Le Chatelier mould** (a split brass cylinder, 30 mm dia. and 30 mm high, with two long indicator arms) with paste of standard consistency, cover with glass plates.
2. Keep it **in water for 24 hours** and measure the distance between the indicator points.

3. **Heat the water to boiling and boil** (1 hour as in the slides; IS 4031 Part 3 specifies 3 hours), then **cool** to the original temperature and measure again.
4. The **increase in distance** between the indicator points is the expansion. IS 269 limits it to **10 mm** for OPC.
5. If the cement fails, **spread it out and aerate it for 7 days** (free lime may hydrate or carbonate in that time) and test again. The second expansion must not exceed **50% of the original limit** (i.e. 5 mm).
6. A cement that fails **both** tests should not be used.

Unsoundness due to free lime is **very rare** in modern cements.

5.3 Autoclave test

Sensitive to **both magnesia and free lime**, so it is the more severe test.

1. Cure a **neat cement paste bar** in humid air for **24 h** and measure its length.
2. Heat it in an **autoclave** with high-pressure steam, raising the pressure to **2 MPa in about 1 hour**.
3. **Hold** the temperature and pressure for **3 hours**.
4. Cool the autoclave so that the **pressure falls within 1.5 hours**.
5. Cool the specimen in water to **23 °C within 15 minutes**; after another 15 minutes **measure its length**.
6. **Expansion must be less than 0.8%** of the original length.

6 Strength of cement

- Strength is tested on **cement–sand mortar** (or concrete) made under controlled conditions; neat paste is not used because it shrinks and cracks.
- Tests can be in **direct tension, compression or flexure**; compression is the standard.
- **Compression test (IS 4031 Part 6): 70.6 mm cubes** (face area 5000 mm^2) of **1 part cement : 3 parts standard sand** by mass, with water = (P/4 + 3)% of the combined mass of cement and sand. The cubes are compacted on a vibration machine, cured, and tested at 3, 7 and 28 days.

Minimum compressive strength, MPa (IS 269:2015):

Age (days)	OPC 33	OPC 43	OPC 43S	OPC 53	OPC 53S
3	16	23	23	27	27
7	22	33	37.5	37	37.5
28	**33**	**43**	43	**53**	53
Maximum at 28 days	48	58	–	–	–

The **grade number is the 28-day strength in MPa** (N/mm^2).

7 Types of cement

7.1 Why there are different types

- Cements with different **chemical (compound) composition** behave differently, e.g. rate of strength gain, heat of hydration and sulfate resistance.
- By changing the **mix of raw materials** (and the fineness), manufacturers make cements with the properties needed.
- Many types were developed specifically to make **concrete more durable** (against heat cracking, sulfates, etc.).

7.2 British and American (ASTM) names

British name	ASTM type
Ordinary Portland	Type I
Modified cement	Type II
Rapid-hardening Portland	Type III
Low-heat Portland	Type IV
Sulphate-resisting Portland	Type V
Portland blast-furnace (slag) cement	Type IS (MS)
Low-heat Portland blast-furnace	–
White Portland	–
Portland-pozzolan	Type IP
(Air-entraining)	Type IA

7.3 Compound composition of ASTM types (%)

Type	C_3S	C_2S	C_3A	C_4AF
I (ordinary)	45–55	20–30	8–12	6–10
II (modified)	40–50	25–35	5–7	10–15
III (rapid hardening)	**50–65**	15–25	8–14	6–10
IV (low heat)	**25–35**	**40–50**	5–7	10–15
V (sulphate resisting)	40–50	25–35	**0–4**	10–12

7.4 Properties compared with Type I

Type	What changes	Result and use
II, modified	Slightly **lower C_3S and C_3A**	**Moderately low heat**; moderate sulfate resistance. Used where moderate heat is wanted or **moderate sulfate attack** may occur
III, rapid hardening	Slightly **higher C_3S** and **higher fineness**	**High early strength**. Allows **early removal of formwork**; protects against **early frost damage** in cold weather
IV, low heat	**Lower C_3S** (lower than Type II) and lower C_3A; more C_2S	**Low heat**, but **slower strength development** than OPC. For **mass concrete** (dams, large foundations)
V, sulphate resisting	**Low C_3A** (0–4%)	**Resists sulfate attack**. For **marine structures**, foundations in sulfate soils; its heat is not much higher than low-heat cement

7.5 Strength development of different types

- Plotted against age on a log scale, the types differ a lot at **early ages**: Type III is strongest early and Type IV weakest.
- In **well-hydrated, mature** paste the differences become **small**; the curves meet at around 2–3 months.
- **A cement that hardens slowly tends to reach a higher ultimate strength** (Types II and IV end up highest after years), because its hydration products form more slowly and pack more densely.

7.6 Blended cements

Cement	Replacement	Notes
Portland blast-furnace slag cement (ASTM Type IS; Indian PSC)	Slag **25–70%** by mass of cement	Ground granulated blast-furnace slag
Portland-pozzolana cement (ASTM Type IP; Indian PPC)	Pozzolana **15–40%** by mass of cement	Usually fly ash; calcined clay also used

Advantages of blended cements:

- **Pozzolanic reaction → additional C-S-H**: the pozzolan reacts with the CH released by cement hydration (CH + S + H → C-S-H).
- **Pore refinement**: the extra C-S-H fills capillary pores, so the concrete is less permeable.
- **Increased durability** (sulfate, chloride and alkali–silica resistance).
- Overall aim: **better strength and durability while using less cement** (lower cost, lower CO_2).

7.7 Indian Standards for cement

Cement	Code listed in the slides	Current version
Ordinary Portland cement	IS 269:1989	IS 269:2015 (covers OPC 33, 43, 53, 43S, 53S)
Low-heat Portland cement	IS 12600:1989	IS 12600:1989
Rapid-hardening Portland cement	IS 8041:1978	IS 8041:1990
Portland pozzolana cement	IS 1489:1976	IS 1489 (Part 1):2015 (fly ash based)
Portland slag cement	IS 445:1976 (*as printed on the slide*)	**IS 455**:2015

> **Code number**
>
> The slides print the slag cement code as "IS 445"; the Indian Standard for Portland slag cement is **IS 455**.

Assignment 1 from the slides

- (a) List the specifications of the different types of cement used in construction, and prepare a table comparing their properties and cost with OPC.
- (b) Merits and demerits of each type.
- (c) Specific applications of each type, with examples.

The tables in section 7 are a good starting point.

Quick revision

- Fineness: finer → faster hydration and strength. Min 225 m^2/kg (OPC 33/43/53), 370 m^2/kg (43S/53S). Blaine (IS 4031-2) measures the time for a fixed volume of air to pass through a cement bed.
- Standard consistency: 10 mm plunger stops 5–7 mm from bottom; usually 26–33% water.
- Initial set: 1 mm needle, 0.85P water, stops 5 mm from bottom; min 30 min (60 min for 43S/53S). Final set: annular attachment fails to mark; max 600 min. Final ≈ 90 + 1.2 × initial.
- False set (remixable, little heat, dehydrated gypsum) vs flash set (no gypsum, big heat).
- Unsoundness: delayed hydration of free lime, MgO, excess gypsum. Le Chatelier (free lime only), limit 10 mm; autoclave (lime + MgO), limit 0.8%.
- Strength: 70.6 mm 1:3 mortar cubes; 28-day strength = grade (33/43/53 MPa).
- Type III: more C_3S + finer → early strength. Type IV: less C_3S, more C_2S → low heat, mass concrete. Type V: $C_3A \leq 4\%$ → sulfate resistance. Blended: slag 25–70%, pozzolana 15–40% → extra C-S-H, pore refinement, durability.

Questions to practise

1. Why is fineness of cement important? Describe the Blaine air permeability test. (6 marks)

2. What is standard consistency of cement? How is it determined? (4 marks)
3. Describe the determination of initial and final setting times with the Vicat apparatus. Why does the code specify a minimum initial and a maximum final setting time? (8 marks)
4. Distinguish between false set and flash set. (3 marks)
5. What is unsoundness of cement? What causes it? Describe the Le Chatelier and autoclave tests. (10 marks)
6. Describe the compressive strength test of cement and give the IS 269 strength requirements for OPC 33, 43 and 53. (5 marks)
7. Compare the compound composition and properties of ASTM Type I, III, IV and V cements and give one use for each. (8 marks)
8. Why does a cement that hardens slowly reach a higher ultimate strength? (2 marks)
9. What are blended cements? State their advantages. (5 marks)